

The Impact of Vertical Scaling Decisions on Growth Interpretations

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Most growth models implicitly assume that test scores have been vertically scaled. What may not be widely appreciated are the different choices that must be made when creating a vertical score scale. In this paper empirical patterns of growth in student achievement are compared as a function of different approaches to creating a vertical scale. Longitudinal item-level data from a standardized reading test are analyzed for two cohorts of students between Grades 3 and 6 and Grades 4 and 7 for the entire state of Colorado from 2003 to 2006. Eight different vertical scales were established on the basis of choices made for three key variables: Item Response Theory modeling approach, linking approach, and ability estimation approach. It is shown that interpretations of empirical growth patterns appear to depend upon the extent to which a vertical scale has been effectively “stretched” or “compressed” by the psychometric decisions made to establish it. While all of the vertical scales considered show patterns of decelerating growth across grade levels, there is little evidence of scale shrinkage.

Keywords: accountability, growth models, item response theory, vertical scaling

The November 2005 announcement of the Growth Model Pilot Program (Spellings, 2005) and the 2007 expansion of the pilot program (Spellings, 2007) have allowed individual student growth trajectories to factor into the educational accountability framework mandated by No Child Left Behind (NCLB) for the first time. The incorporation of student growth makes the evaluation of change in student achievement an important complement to an evaluation of achievement “status” at any single point in time. However, there are thorny issues to be resolved in the marriage of status and growth. Some of these issues are political and even philosophical. For instance, are all schools and their students being held to standards that are

reasonable for them to attain? How does one properly distinguish growth as “high” or “low”? To what extent and for how long does high growth make up for low status? Should schools with low growth and high status be sanctioned? These are questions with which policy-makers will need to grapple in the reauthorization of NCLB; however, in the present paper we direct our attention toward a foundational measurement issue that is typically posed behind the scenes by psychometricians who develop large-scale assessments: How can a vertical score scale be created to best facilitate test score comparisons across grades over time? As we will demonstrate empirically, there is no single “correct” answer to this seemingly esoteric question, but the decisions made

in the process of creating a vertical scale can have important repercussions for subsequent growth interpretations. As Tong and Kolen (2007) point out:

No consensus exists in the literature as to which set of procedures produces a vertical scale that most adequately captures the nature of students’ growth . . . generally each test developer determines the procedures used in the development of its own scale. (p. 228)

Before we proceed, it is important to note that in the present study our focus is first and foremost on the measurement properties of a longitudinal score scale in the context of *growth models*, rather than in the context of *value-added models*. The easiest way to define and distinguish features of these models is by example. The simplest growth model is one that compares the difference in test scores from 1 year to the next. In such a model, achievement is always on the left-hand side of the equation—a student’s achievement in a current year never depends on achievement in a prior year. An extension of this approach to more than two time points would encompass various specifications of hierarchical linear models (c.f., Raudenbush & Bryk, 2001; Singer & Willett, 2003). In these models, student-level growth trajectories are the principal phenomenon of interest and are thus almost always parameterized. The most well-known example of a value-added model is the “layered

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model” (Ballou, Sanders, & Wright, 2004; Sanders, Saxton, & Horn, 1997) that underlies the Educational Value-Added Assessment System (EVAAS). In this model, which has been used for a number of years in states such as Tennessee and Pennsylvania, there is no attempt to formally parameterize student growth; rather, the goal is to estimate a teacher-specific residual that can be interpreted as the “value-added” to student achievement by a given teacher in a given grade.

A key distinction between growth models and value-added models is the way they are used and interpreted within systems of educational accountability. The simplest use of a growth model is to summarize and/or project trends in student achievement over time. In most cases these trends are given a criterion-referenced interpretation to evaluate how much growth is considered adequate or inadequate. In contrast, value added models are used to estimate quantities that support causal inferences about the specific contributions that teachers and/or schools make to student achievement in any given year. These quantities take on a norm-referenced interpretation as the value added by a given teacher or school relative to some empirically defined quantity such as the statewide average.¹

An implicit assumption underlying all growth models is that the test scores for a given subject (i.e., reading, math, etc.) in earlier grades can be compared to the test scores in later grades along some absolute scale. On the surface, observed performance on such tests (i.e., the proportion of items answered correctly) is clearly not comparable because the tests will necessarily differ with respect to their difficulty. For example, in an absolute sense a reading test in Grade 3 will be easier than a reading test in Grade 4, which will be easier than a test in Grade 5, and so on. To place student performance onto a common scale, scores from two or more tests must be linked statistically to create what is known as a vertical scale.

The purpose of this article is to compare the empirical patterns of growth that result as a function of different approaches to creating a vertical scale. Two notable aspects of the empirical analyses in this study are the use of item-level data over a longitudinal time span, and the linking of tests using both dichotomous and polytomous items. Our basic strategy is to use item

response theory (IRT: Lord, 1980; Lord & Novick, 1968) to create different vertical scales, but to vary the decisions made at three critical junctures of the scaling process. Combinations of these decisions lead to eight different vertical scales. The central outcomes of interest are potential differences in means and standard deviations (*SDs*) among the vertical scales from year to year and how these outcomes compare to previous empirical studies of IRT-based vertical scaling. We focus on IRT-based approaches to vertical scaling because this is the predominant method employed by commercial test developers.

In the following section, we provide an overview of the stages through which an IRT approach is used to establish a vertical score scale. Next, we briefly review previous studies that have compared and contrasted a variety of different vertical scaling approaches to motivate the “decision factors” that we manipulate in the present study. In the third section, we describe the data and design for our procedures, and in the fourth section we present the results. The article concludes with a discussion of our findings and the implications for educational accountability systems that hinge upon growth interpretations along a vertical scale.

Using Item Response Theory to Establish a Vertical Scale

Choosing an IRT Model

In IRT, an examinee’s score on a test item is modeled probabilistically as a function of the examinee’s latent proficiency and an item’s characteristics. Let the variable X_{pi} represent the response of examinee p to item i . Given a test consisting of multiple-choice (MC) items, $X_{pi} = 1$ for a correct-item (CI) response, and $X_{pi} = 0$ for an incorrect response. The item characteristic curve (ICC) for what is known as the three-parameter logistic model (3PLM: Birnbaum, 1968) can be written in the following form

$$P(X_{pi} = 1 | \theta_p, \alpha_i, \beta_i, \gamma_i) = \gamma_i + (1 - \gamma_i) \times \frac{\exp(D\alpha_i(\theta_p - \beta_i))}{1 + \exp(D\alpha_i(\theta_p - \beta_i))}. \quad (1)$$

In the formulation above, θ_p represents latent proficiency (measured in logits, which are the log of the odds of a CI response), α_i represents item discrimina-

tion (slope of the ICC at the location of item difficulty), β_i represents item difficulty (value of θ_p at the ICC inflection point), γ_i represents a lower asymptote (pseudo-guessing parameter), D represents a scaling constant that equals 1.7, and “exp” represents the exponential function. When constraints are placed on the guessing and discrimination parameters in the 3PLM, two other commonly used models for dichotomously scored items can be derived. For the two-parameter logistic model (2PLM: Birnbaum, 1968) the guessing parameter γ_i is constrained to zero, and for the one-parameter model (1PLM) the 2PLM is constrained so that the discrimination parameters α_i are the same for all items. Though it has different historical and philosophical origins, mathematically the Rasch model (Rasch, 1960) can be viewed as a special case of the 1PLM where all of the item discriminations are constrained to equal one.

In an analogous fashion, constructed-response (CR) items, generally referred to as polytomous items, can be modeled with what is known as the generalized partial credit model (GPCM: Muraki, 1992). The GPCM is akin to the 2PLM in a polytomous context (there is generally no guessing associated with constructed response items), and in the same way that the 2PLM can be constrained to produce the 1PLM, the discrimination parameters can be constrained to be equal for all polytomous items. This is known as the partial credit model (PCM: Masters, 1982). In practice, when tests for large-scale assessment include a mixture of dichotomous and polytomous items they are commonly modeled using either a combination of the 3PLM and GPCM or the 1PLM and PCM.

One fundamental assumption all IRT models make is that of local independence, which posits that conditional on an examinee’s latent proficiency, θ_p , item responses within a given test should be statistically independent observations. A related assumption is that the construct of measurement is unidimensional. Given that these assumptions are met and that the correct functional form has been specified, all IRT models share a very useful property: parameter invariance. According to this property, which is the linchpin for using IRT to establish a vertical score scale, item parameters are independent of the specific characteristics of the sample

of test-takers used to estimate them. This implies, for example, that if fourth grade students and fifth grade students answer the same items on a reading test, the difficulty of the items should be the same regardless of which group was used to estimate them—even though in an absolute sense the items should generally be easier for the fifth grade students because they will likely have higher reading ability than fourth grade students.

Using IRT, two test score scales can be linked together provided that (a) the tests measure the same construct, and (b) the tests share a set of common items. This strategy constitutes what is formally known as a “common-item nonequivalent groups” (Kolen & Brennan, 2004), or a “nonequivalent groups with anchor test” (von Davier, Holland, & Thayer, 2004) linking design. The item response data we use in the present study are based upon this design.²

Choosing a Linking Approach

In general, there are two IRT approaches used to create a vertical scale across two or more different grades: separate or concurrent linking.³ Under a separate approach, item parameters and examinee ability (i.e., the achievement level that corresponds to an individual’s scale score) are first estimated in separate runs for each grade-level test using the same IRT model and ability estimation approach (described below). Ultimately, the aim is to place the ability estimates from the various tests onto a common scale, and under the separate linking approach this is accomplished using a set of linear transformations. For example, imagine that we administer two reading tests at the end of a school year, one to Grade 4 students and, the other to Grade 5 students. Say also that we are interested in placing the scores for fifth grade students on the fourth grade scale. The following transformation

$$\theta_{p5}^* = A\theta_{p5} + B$$

can be used, where A and B are linking constants that adjust the SD and mean ability of the fifth grade scores, respectively, and θ_{p5}^* is the transformed score. If the same examinees take both tests, the linking constants can be readily computed, but, as noted earlier, such a design is not generally feasible. Within a common-item nonequivalent groups

linking design, these constants can be found through a comparison of the differences between the estimated parameters for a set of common items on the two tests. Continuing with this illustration, say each test has 50 MC items; 15 items are common to both grades, and in general, the items on the Grade 5 test are more difficult than items on the Grade 4 test. The item parameters for 100 items (50 per grade) could be estimated with a 3PLM. However, only the information about the 15 common items would be used to link the two tests. Given the IRT property of parameter invariance, the discrimination, difficulty, and guessing parameter of each common item ($i = 1, \dots, 15$) on the Grade 5 test ($\alpha_{i5}, \beta_{i5}, \gamma_{i5}$) will have the following relationship with the corresponding parameters for the same items on the Grade 4 test ($\alpha_{i4}, \beta_{i4}, \gamma_{i4}$):

$$\begin{aligned}\alpha_{i4} &= \frac{\alpha_{i5}}{A}, \\ \beta_{i4} &= A\beta_{i5} + B, \\ \gamma_{i4} &= \gamma_{i5}.\end{aligned}$$

Again, A and B are the linking constants used for the linear transformation. Because of the relationship between the item parameters and θ , only one set of linking constants is needed to transform all of the item parameters to the Grade 4 scale. Conceptually, the A constant adjusts the discrimination of the items and the B constant adjusts the difficulty. Note that there is no change to the guessing parameters. While the equations above capture the theoretical relationship between the common items on the two tests, the linking constants are still unknown. Various methods have been proposed to estimate these constants, but the Stocking-Lord method (Stocking & Lord, 1983), which minimizes the difference between test characteristic curves represented by the common items between grades, is most commonly used.

When there is a need to place scores from more than two tests on a common scale, the linear transformation presented earlier can be extended using a process known as “chain linking.” For instance, if there are common items between Grades 4 and 5, and a separate set of common items between Grades 5 and 6, linking constants can first be estimated for each grade pair (C_{45} and C_{56}). The Grade 5 scores would be transformed to the Grade 4 scale—the base grade—using the C_{45} constants, and the Grade 6 scores would

be transformed to the Grade 4 scale by first transforming them to the Grade 5 scale using the C_{56} constants, and then the C_{45} constants.

In contrast to the separate linking approach described earlier, under concurrent linking, all item parameters for all grades are estimated in one step during which different underlying population ability distributions are specified for each group of students taking the tests across grades (Bock & Zimowski, 1997). With the differences in ability for the two populations of test-takers taken into account directly, the parameters of items common to both groups serve to anchor the score scale. If this approach were used in the aforementioned hypothetical example, 85 sets of item parameters would be estimated: 70 for the unique items and 15 for the common items. All 85 items are automatically calibrated to be on the same scale with the 15 common items providing the link between the two tests.

Choosing an Ability Estimator

Once item parameters have been estimated (whether separately or concurrently), a specific scale score can then be estimated for each individual. The two most common approaches for accomplishing this are maximum likelihood (ML) estimation and expected *a posteriori* (EAP) estimation (Bock & Aitkin, 1981). With the ML approach, the joint probability of an examinee’s response pattern is maximized to determine the most likely ability level. With the EAP approach, the joint probability distribution is weighted by a set of quadrature points—typically associated with a normal distribution—to provide an estimate of ability. The key tradeoff between these two methods is one of efficiency versus bias. ML estimates are asymptotically consistent, but they can be inefficient for examinees with ability near the tails of the distribution. EAP estimates are biased, but they are easily calculated, and as Bock and Mislevy (1982, p. 439) noted “the EAP estimator has minimum mean square error over the population of ability and, in terms of average accuracy, cannot be improved upon.” The estimation of a student-specific scale scores by grade is the last step before it becomes possible to make comparable interpretations of longitudinal growth trajectories.

A Brief Review of the Literature on IRT-Based Vertical Scaling

Research on the use of IRT to create a vertical score scale (sometimes referred to as a “developmental” scale) dates back to the 1970s. Much of the earliest research focused on the use of the 1PLM to link tests that differed in difficulty (Divgi, 1981; Gustafsson, 1979; Holmes, 1982; Loyd & Hoover, 1980; Slinde & Linn, 1978, 1979a, 1979b; Wright, 1997). Subsequent research compared the use of both the 1PLM and the 3PLM to more traditional approaches (i.e., equipercentile methods, Thurstone scaling) (Kolen, 1981; Marco, Petersen, & Stewart, 1983; Petersen, Cook, & Stocking, 1983; Skaggs & Lissitz, 1986). By the mid 1980s, these comparisons had led to one rather puzzling finding: IRT-based vertical scales indicated that within-grade variability in test performance decreased over time, a phenomenon described as “scale shrinkage.” This was in contrast to the findings from traditional scaling approaches, which generally showed the opposite pattern. Camilli (1987), Camilli, Yamamoto, and Wang (1993), and Williams, Pommerich, and Thissen (1998) speculated that the decreasing variability could be explained by problems associated with IRT ability estimation for low and high-scoring students across grade level tests. The latter may have been due to the extensive use of the IRT software LOGIST (Wingersky, Barton, & Lord, 1982) and BICAL (Wright & Mead, 1978). Both programs depend upon joint maximum likelihood (JML) estimation of item and person parameters, and the use of JML has been shown to lead to bias in these estimates under certain conditions. (For details, see Baker & Kim, 2004.) Kolen (2006), notes that among studies using marginal maximum likelihood (MML) estimation (Bock & Aitken, 1981), there has been no consistent pattern of decreasing within-grade variability. When a pattern of decreasing variability does occur, one might reasonably argue on the basis of research by Yen (1985) that a violation of the IRT assumption of unidimensionality is a likely culprit.

If anything, the research interest in IRT-based vertical scaling has increased since the passage of NCLB (c.f., Tong & Kolen, 2007). Our review of this research literature has led us to focus on three factors which we will be varying in the process of creating a vertical

scale: (1) the specification of the underlying IRT model, (2) whether linking is done separately or concurrently, and (3) whether proficiency estimation is based up on a ML or EAP approach. Below we explain the rationale behind our manipulation of each factor, and to what extent we can predict the marginal impact each factor should have on growth interpretations.

What IRT Model Should be Specified?

The decision about which IRT model to use when scaling a standardized test may be made for statistical, pragmatic, and/or even philosophical reasons.⁴ From a statistical perspective, more complex models such as the 3PLM and GPCM will always fit the data better than more parsimonious models such as the 1PLM and PCM, and will provide for more precise estimates of examinee proficiency. On the other hand, if there is an interest in developing a score scale with interval properties, then at least in theory this is possible using a combination of the 1PLM and GPCM; it is much less plausible when using the 3PLM (see Ballou, 2009, for details). Finally, more parsimonious models tend to lend themselves to more transparent interpretations. The 3PLM and GPCM weight items relative to their discrimination parameters, whereas item weight is constant under the 1PLM and PCM. As such, provided there is no missing data, the rank orders of student raw scores and 1PLM/PCM scores will be the same. The same is generally not true for the 3PLM and GPCM.⁵

When IRT models are used to craft a vertical scale, it appears that the more complex specification has generally out-performed the parsimonious one. In their review of the literature, Skaggs and Lissitz (1986) concluded that use of the 1PLM has been shown to produce “inadequate results” and counseled practitioners toward cautious use of the 3PLM instead. The main problem found with use of the 1PLM in vertical scaling appears to have been a failure to account for guessing when the scale score of examinees with low proficiency are linked on the basis of items that are very difficult for them to answer.

We have chosen to create vertical scales using the 1PLM and PCM as a contrast to the use of the 3PLM and GPCM for two reasons in particular. First, in practice, many states and/or school districts do in fact use the 1PLM

and PCM to scale their test scores. Hence, it is important to demonstrate the impact this choice will have on growth interpretations relative to the use of a more complex IRT model. Second, many of the studies that have been pessimistic about the use of the 1PLM for purposes of vertical scaling were either using tests with items that did not fit the model (Slinde & Linn, 1978; Divgi, 1981) or were estimating both item parameters and student proficiency using JML. In the present study, we are using a test in which the fit of the items to the 1PLM appears relatively good, and all parameter estimation is based on MML techniques. One consequence that we can anticipate when comparing grade-by-grade growth from the 1PLM/PCM to the 3PLM/GPCM is that with the latter the score scale will be expanded relative to the former, for reasons we describe in the next section. Hence in an absolute sense, growth should appear more impressive at first glance using the 3PLM/GPCM.

Separate or Concurrent Linking?

Separate and concurrent linking approaches each have strengths and weaknesses. The separate approach is easy to implement, but because it is unlikely that linking constants are estimated without error, to some extent this additional error will contaminate the transformed scale. As Kim, Lee, & Kim (2008) have shown, this will particularly be the case as the transformed grade departs further from the base grade. In contrast, with the concurrent approach only one model must be specified to estimate all the parameters and create the vertical scale. In this regard there is no comparable source for “linking error,” unless one were to conceptualize such error with respect to the choice of common items in the underlying linking design (Michaelides & Haertel, 2004). In practice, testing companies have been more likely to build and maintain a vertical scale using a separate linking approach. There are at least two reasons for this. First, because testing companies typically employ large item banks, it is often not feasible to estimate parameters for all items simultaneously when new items are added. Second, and probably more importantly, when the definition of the measured construct changes across grades, concurrent estimation can introduce bias throughout

the entire scale because the assumption of unidimensionality has been violated, whereas separate linking may mitigate such bias by relying only on pairwise linking across grades (Béguin & Hanson, 2001; Béguin, Hanson, & Glas, 2000).

The debate over the use of separate versus concurrent approaches is relatively new to the research literature, and at present there is no consensus—either theoretically or empirically—as to which should be preferred (Ito, Sykes, & Yao, 2008; Hanson & Béguin, 2002; Kim & Cohen, 1998). In the present study, we develop vertical scales using a separate approach, and a modification of a purely concurrent approach. In most applications of IRT-based vertical scaling, a cross-section of examinees in a single year are tested across a vertical grade span, and this becomes the basis for establishing a vertical link across the tests. Instead, as we describe in more detail in the next section, our data comes from two longitudinal cohorts of students—the same students are included in the data set on multiple occasions. This created violations of the IRT local independence assumption that we were unable to resolve with the multigroup IRT software at our disposal. For this reason, instead of using a purely concurrent approach, we used a hybrid method similar to the one taken by Karkee, Lewis, Hoskens, Yao, & Haug (2003) in which item parameters for the within-grade/ across-year assessments were estimated concurrently, and then separate calibration was used to create the across-grade links. If growth interpretations differ by calibration approach, the most likely explanation is linking error, multidimensionality, or some combination of the two.

Ability Estimation: ML or EAP?

The choice of ability estimator is also relatively new to the research literature, but it is a factor flagged by both Kolen & Brennan (2004) and Kolen (2006) as an important distinguishing feature among vertical scales. For states scaling their tests with the 1PLM and PCM with the software Winsteps (Linacre & Wright, 1998), a choice of ML estimation is implicit. For states scaling their tests with the 3PLM and GPCM, either ML or EAP estimation could be used. The impact of this choice will have a predictable impact

on growth interpretations: the use of EAP estimates will contract the scale relative to the use of ML estimates. Because EAP estimates are shrunken to a population mean, they will be less variable than ML estimates. On the other hand, when aggregated across students and schools at the state level, both EAP and ML estimates should have the same population mean.

Methods

Data

The data we use in this study are longitudinal item responses from the Colorado Student Assessment Program (CSAP) reading test. The CSAP vertical scale is based on a common item nonequivalent groups linking design that was established by the state's test contractor, CTB/McGraw-Hill, in 2001. The vertical scale was created by scaling each grade-specific test from 3 to 10 with the 3PLM and GPCM, linking the tests together using a separate approach (with Grade 7 as the base grade of the scale), and then producing ML ability estimates for student-level scores. Since its creation in 2001, parallel test forms have been administered at each grade. Item parameters and ability estimates from subsequent tests are horizontally equated so that they can be compared relative to the 2001 scale (i.e., the base scale). To maintain the vertical scale within this linking framework, there is no single set of common items across years in each grade; rather, different sets of items are shared across different years, and there are a limited number of across-grade common items in any given year.

The vertical scales created in the present study derive from longitudinal data we obtained from the Colorado Department of Education for two cohorts of students. The first cohort included students who were in Grade 3 in 2003 and Grade 6 in 2006; the second cohort included students who were in Grade 4 in 2003 and Grade 7 in 2006. Our vertical scaling design involved the linking of tests for adjacent grades in the same year, as well as the linking of tests for the same grade in adjacent years. This design is summarized in Table 1 further. The CSAP reading tests used to create different vertical scales contained a mix of MC and CR items. In Grade 3 the test consisted of 34 MC items and 7 CR items; in Grades 4–7 the respective numbers were about 70 MC items and 14 CR items. The number of common MC and CR items across adjacent grades or years ranged from 7 to 20 and 0 to 4, respectively. Each of the eight grade by year combinations (Grades 3 to 7 in 2003–2006) used for the analysis included an average of 55,681 students enrolled in 1,379 unique public schools. Roughly 64% of the students self-identified as white, 26% as Hispanic, 6.2% as black, 3% as Asian/Pacific Islander, and 1.3% as Native American.

While the design we use to create our vertical scales—common item nonequivalent groups—is consistent with the design underlying the CSAP operational scale, it differs with respect to (a) our choice of base year and grade (2003 and 3, respectively), and (b) the number of common items available to link tests across grades or across years. As a rule of thumb, Kolen & Brennan (2004, p. 271) have recommended that

Table 1. Unique and Common Items on CSAP Reading Test by Grade and Year

Grade	Year			
	2003	2004	2005	2006
3	(34, 7) (13, 3)			
4	(56, 14)	(15, 3) (56, 14) (9, 3)		
5		(56, 14)	(20, 2) (58, 14) (11, 4)	
6			(57, 14)	(7, 0) (57, 14) (10, 4)
7				(58, 14)

Note: First value in parenthesis represents number of MC items, second value represents number of CR items. Values in bold represent common items.

Table 2. IRT-Based Vertical Scaling Models

Item Response Model		Linking Approach	
		Separate	Hybrid
EAP Scale Scores	3PLM/GPCM	1	2
	1PLM/PCM	3	4
ML Scale Scores	3PLM/GPCM	5	6
	1PLM/PCM	7	8

the number of common items available for test linking equal roughly 20% of the total number of items on any single test form. In our design there is one link that clearly falls short of this rule of thumb: for the Grade 6 tests between 2005 and 2006 there are just seven common items (about 10% of the total items). Given this limitation in our design, inferences about scale score growth that involve Grade 6 test scores should be made with caution.

Creating Vertical Scales

We created eight vertical scales that differ with regard to three factors: (1) the IRT model used to estimate item parameters, (2) the linking method used to place the parameters from different

grades onto a common scale, and (3) the method used to estimate student-level scale scores. Table 2 provides an overview of the eight scales that result from the combination of these variables. The operational CSAP vertical scale for reading is currently based on the combination of factors represented in cell number 5: 3PLM/GPCM, separate linking, and ML score estimation. However, as discussed in our review of the literature, the scales represented by any of the other cells would be defensible alternatives.

The two different linking approaches taken in this study are illustrated graphically in Figures 1 and 2. In Figure 1, each oval represents a separate linking of tests across grades in the same year (vertical direction), or across years in the same grade (horizontal direction). Under the hybrid approach illustrated in Figure 2, we separately linked tests across grades in the same year (ovals), but concurrently linked tests across two years in the same grade (rectangles). When using the separate approach, we first estimated item parameters for each grade estimated independently. Next, using the Stocking-Lord method (Stocking & Lord, 1983), we estimated linking constants for the various within-year/across-grade and within-grade/across-year pairs illustrated in Figure 1. We used these constants to transform both item parameters and estimates of latent proficiency onto the Grade 3 scale using chain linking. Linking constants and the chain linking transformation were computed using the R package *plink* (Weeks, 2007). Under the hybrid approach, we first estimated item parameters for each of the within-grade/across-year assessments concurrently. Next, we used the Stocking-Lord method to estimate linking constants for the across-grade linkages shown in Figure 2. The item parameters and proficiency estimates for these scales were then placed onto the

Grade 3 scale using chain linking. All of the item parameters were estimated using an MML approach in the IRT Command Language program (ICL; Hanson, 2002). We report scores for each vertical scale in logit units.

Results

The means and *SDs* for each of the eight vertical scales we created are summarized in Figures 3 and 4 for the first of our two longitudinal student cohorts:⁶ those students who were in Grade 3 in 2003 and Grade 6 in 2006. For each of these two statistics, there are three “main effects” of interest:

1. The difference between IRT models used to estimate the item parameters (1PLM/PCM vs. 3PLM/GPCM). All scales that involved the 1PLM/PCM combination are denoted graphically by lightly shaded lines, while all scales that involved the 3PLM/GPCM are denoted by darkly shaded lines.
2. The difference between approaches used to link the vertical scales (separate vs. hybrid). All scales that were created using the separate approach exclusively are denoted graphically by solid lines, and the scales created using the hybrid approach are denoted with dotted lines.
3. The difference between approaches used to estimate student-level scale scores (EAP vs. ML). All scales that were created with EAP estimation are denoted graphically with X markers, and the scales created with ML estimation are denoted with O markers.

There are clear patterns in the changes in means and *SDs* for the different vertical scale across grades. We start by examining the different trends in growth by scale shown in Figure 3. For each scale, the growth in score means from grade to grade appears somewhat nonlinear, and decelerating over time. This is consistent with previous findings in the literature (Kolen, 2006, p. 178). The apparent magnitude of growth along the logit scale differs substantially as a function of the underlying IRT model and linking approach. Scales created using the 3PLM/GPCM and separate linking combination give the impression of the most growth; scales created using the 1PLM/PCM and either separate or hybrid linking give

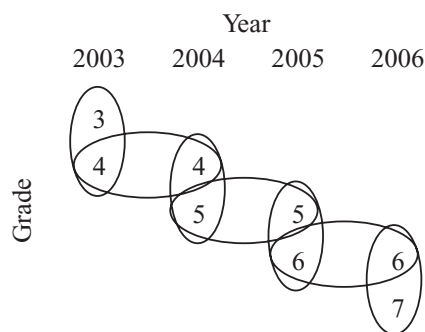


FIGURE 1. Separate linking approach.

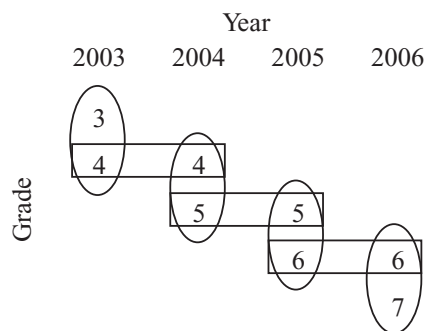
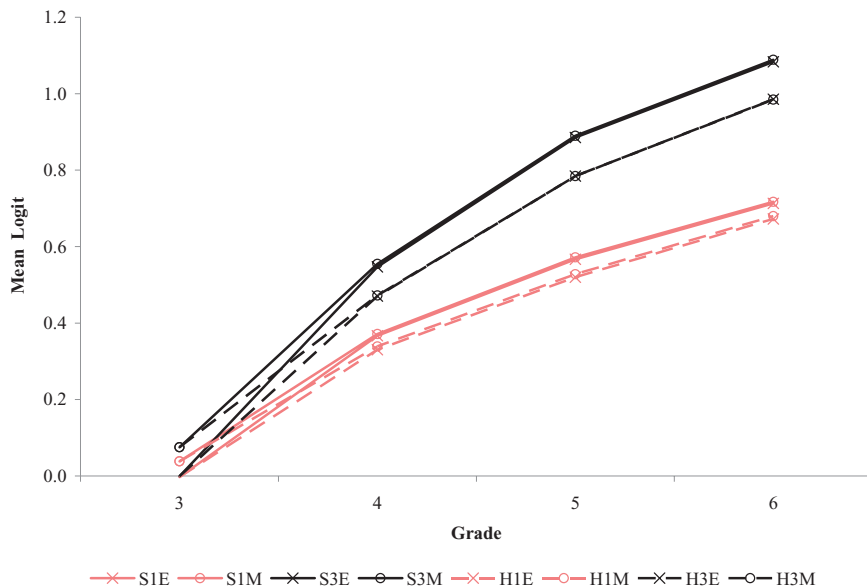
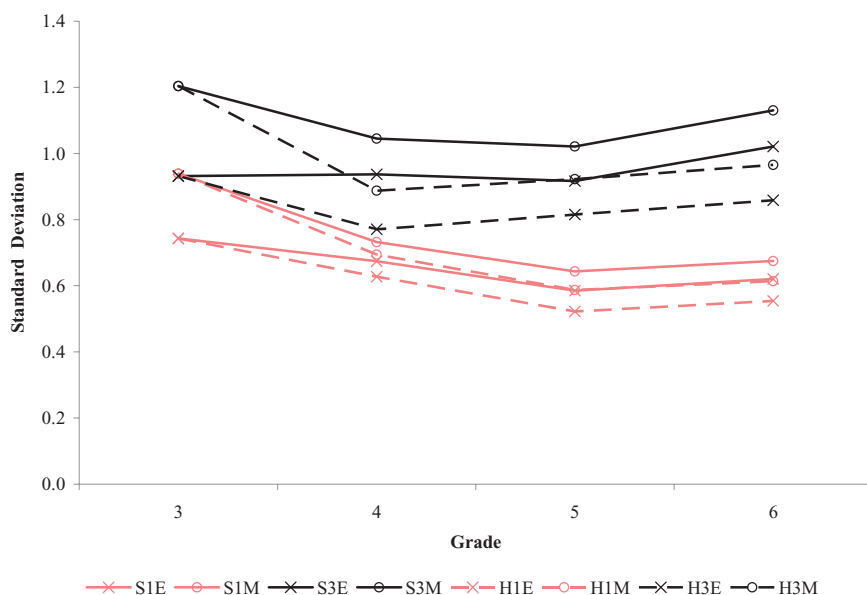


FIGURE 2. Hybrid linking approach.



Note: In the figure above, solid line = separate linking; dashed line = hybrid linking; light line = use of the 1PLM/PCM; dark line = use of 3PLM/GPCM; X marker = EAP estimation; O marker = ML estimation.

FIGURE 3. Growth along CSAP reading score scale from 2003 to 2006.



Note: In the figure above, solid line = separate linking; dashed line = hybrid linking; light line = use of the 1PLM/PCM; dark line = use of 3PLM/GPCM; X marker = EAP estimation; O marker = ML estimation.

FIGURE 4. Variability of CSAP reading score scale from 2003 to 2006.

the impression of the least growth. This was to be expected. The use of the 3PLM/GPCM stretches the score scale because distinct scores are computed for each unique item response pattern. This is in contrast to the 1PLM/PCM, where an examinee's total score serves as a sufficient statistic for his or her scale score; distinct scale scores are computed for each raw score. What was not necessarily expected was that when

3PLM/GPCM estimates are linked using the hybrid approach rather than the separate approach, the observed growth trajectory is shifted downward.

Figure 4 plots the within-grade variability over time for each scale. The first thing to note is that the pattern from grade to grade is inconsistent. From Grade 3 to 4, score variability decreases significantly across all but one scale, whereas from Grade 5 to 6

it stays roughly constant or increases slightly across all eight scales. Between Grades 4 and 5, variability increases for the two scales based on the 3PLM/GPCM that use hybrid linking with ML or EAP estimation, but decreases for all other scales. In general, using the 3PLM/GPCM produces scales with greater variability than the 1PLM/PCM, the use of ML estimation increases the variability in a scale relative to EAP estimation, and the use of hybrid linking appears to decrease scale variability relative to separate calibration. So while the choice of IRT model clearly has the largest impact on scale variability, the choice of linking and ability estimation approach still have a significant impact, especially when combined with use of the 3PLM/GPCM.

One message conveyed by viewing Figure 3 in conjunction with Figure 4 is that interpretations of grade-by-grade growth along a vertical score scale can be misleading unless one also takes into account the associated variability of the scale. This is illustrated in Figure 5, which for illustrative purposes contrasts two extremes: a vertical scale based on the 3PLM/GPCM, separate linking, and ML estimation (the current approach taken in Colorado); and a vertical scale based on the 1PLM/PCM, hybrid linking, and EAP estimation. In terms of the three decisions used to establish each vertical scale in each case, the former scale maximizes score variability while the latter scale minimizes it.

Year-to-year growth along a given scale can be compared while adjusting for the variability of the underlying scale by standardizing the mean differences. Yen (1986) defined an effect size statistic for these purposes as

$$\text{effect size} = \frac{\bar{\theta}_{\text{upper}} - \bar{\theta}_{\text{lower}}}{\sqrt{\frac{\sigma_{\text{upper}}^2 + \sigma_{\text{lower}}^2}{2}}},$$

where $\bar{\theta}_{\text{upper}}$ and $\bar{\theta}_{\text{lower}}$ represent the mean scale scores for the higher and lower grades or years in the scale, respectively, and σ_{upper}^2 and σ_{lower}^2 represent the respective variance for the scores in each grade or year. All else held constant, it will always be the case that effect sizes based on EAP proficiency estimates will be larger than those based on ML proficiency estimates.

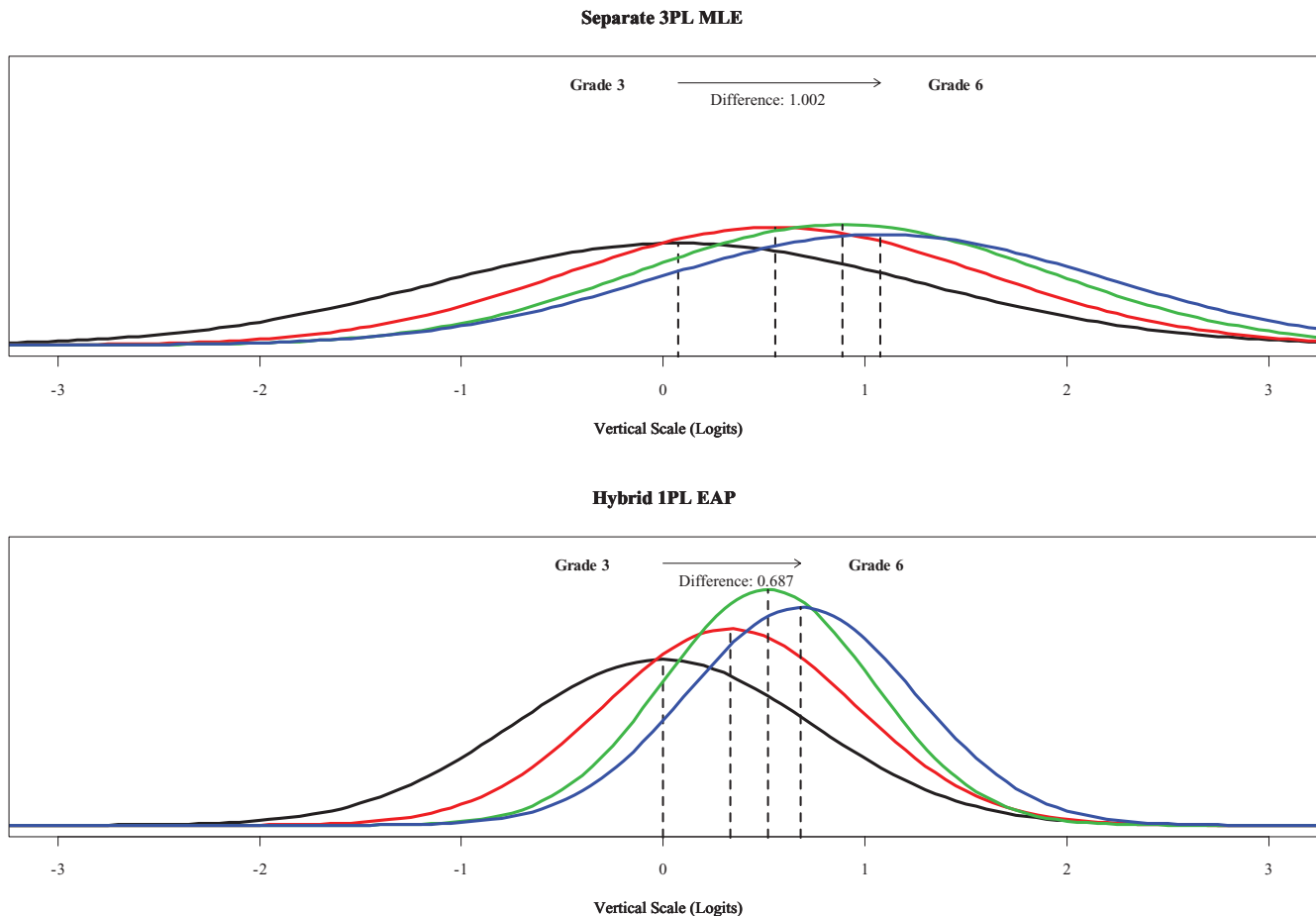


FIGURE 5. Comparing extremes in vertical scales.

The effect size estimates that correspond to the growth from Grades 3 to 4, 4 to 5, and 5 to 6 are shown in Figures 6 and 7 for the four 1PLM/PCM scales and 3PLM/GPCM scales, respectively. In general, these effect size patterns are fairly similar; however, for each across grade comparison, the effect sizes based on two different scales can differ by as much as 10–20% of the average SD across grades. We note that though there are considerable differences between the effect sizes for growth from Grade 3 to 4 as a function of ability estimation approach, these differences are negligible by the time effect sizes are compared for growth from Grade 5 to 6. We have no explanation for this finding at the present time. Finally, we note that the use of the 3PLM/GPCM does not always result in the largest effect sizes. When the latter is used in conjunction with ML estimation, the resulting growth can be anywhere from 5 to 10% smaller in effect size units relative to a vertical scale based upon the 1PLM/PCM and EAP estimation.

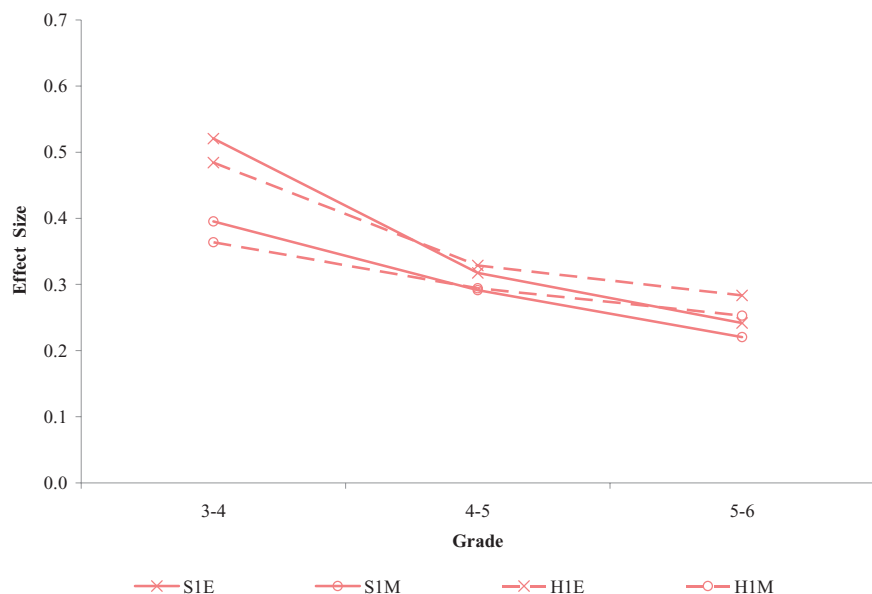
Discussion

Over the years the point has been made repeatedly that when it comes to interpretations of longitudinal growth in student achievement, the metric matters (Seltzer, Frank, & Bryk, 1994; Tong & Kolen, 2007; Yen, 1986). Our findings echo this sentiment. Depending upon the underlying IRT model, linking method, and ability estimation approach, a vertical scale can be effectively stretched or compressed. It follows from this that if one only interprets growth without taking the variability of the scale into consideration, one can be easily misled one way or another. When the scale is interpreted in effect size units such that information about mean changes and scale variability are combined into a single statistic, growth patterns are much more similar, though the results presented here still suggest significant differences in magnitude across some scales—as much as 20% of a *SD*. In other words, even when considering the same item responses on the same tests from the same populations of students, absolute interpreta-

tions of growth in reading achievement can be influenced by the way the underlying scale has been established.

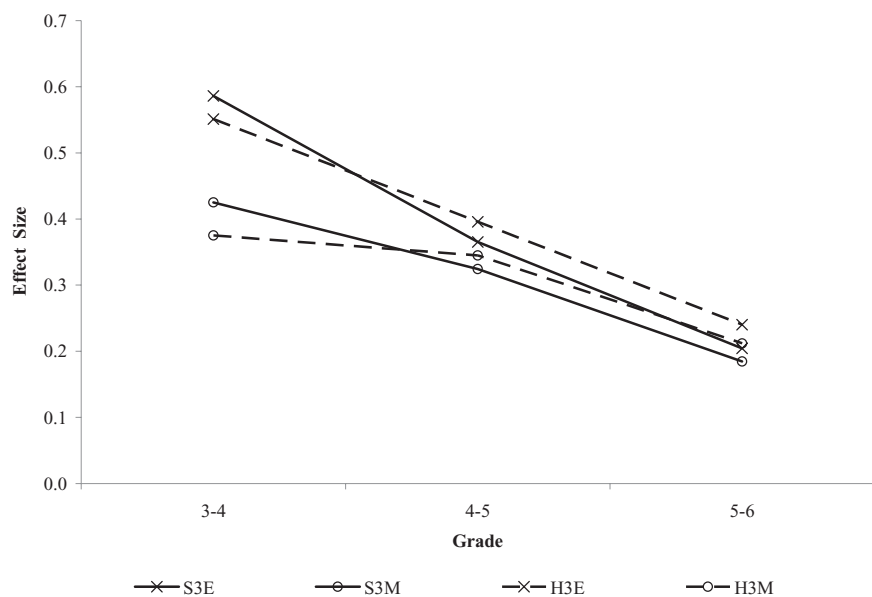
The phenomenon of scale shrinkage was thought to be associated with IRT-based vertical scaling in the early 1980s. Recently, Tong and Kolen (2007) found evidence of severe scale shrinkage for the ITBSs 1992 reading tests for Grades 3–8 when using a 3PLM separate calibration approach with both EAP and ML estimation. In contrast, Ballou (2009) found no such evidence using data from the NWEAs 2005 reading tests for Grades 2–9. The NWEA tests had been vertically scaled using the Rasch model with concurrent calibration.⁷ Our findings do not indicate that a clear pattern of scale shrinkage is associated with any of our IRT-based vertical scaling approaches. Unfortunately, without a deeper understanding of the content differences between the ITBS and CSAP reading tests, we cannot at present reconcile the differences between our findings and those of Tong and Kolen.

What are the ramifications of our findings for uses of vertical scales



Note: In the figure above, solid line = separate linking; dashed line = hybrid linking; X marker = EAP estimation; O Marker = ML estimation.

FIGURE 6. Effect sizes of growth for cohort 1 by 1PLM/PCM scales.



Note: In the figure above, solid line = separate linking; dashed line = hybrid linking; X marker = EAP estimation; O marker = ML estimation.

FIGURE 7. Effect sizes of growth for cohort 1 by 1PLM/PCM scales.

in systems of educational accountability? Even in states with tests that have been vertically scaled, under NCLB scale scores are ultimately converted into discrete performance categories (i.e., “proficient”, “advanced”, etc.) by establishing cut-points on the score scale across grade levels through the process of standard-setting. These proficiency classifications must also be made vertically

comparable (Lewis & Haug, 2005). Whether it is possible to establish “vertically moderated standards” (Lissitz & Huynh, 2003) through what is clearly a subjective process is an open question. One might suspect that there is an interaction between this process and the properties of the underlying vertical scale, but this is pure speculation. In any case, it would seem that as long as states remain consistent in the

approach used to establish and maintain their vertical scales, and likewise in their approach to turn thresholds along this scale into vertically moderated standards, accountability conclusions based upon the sorts of models proposed and approved (so far) under the Growth Model Pilot Program should remain unaffected by the specific vertical scaling approach that has been employed.

In some cases, however, states or school districts may be tempted to move from the use of a growth model as a criterion-referenced means of summarizing and projecting student achievement, to the use of such a model to estimate the norm-referenced value-added “effects” of teacher or schools on student achievement.⁸ This is an example in which a growth model that begins with a primary focus on student achievement trajectories is subsequently parameterized as a value-added model. If schools were to be ranked according to the value-added estimates taken from such a model, to what extent would such rankings be sensitive to the choice of vertical scale? To explore this, we applied an unconditional three-level (time, student, and school) hierarchical linear model (Raudenbush & Bryk, 2002) to the data for 992 Colorado elementary schools as a function of our eight vertical scales. We specified a linear growth function at both the student and school level with random effects at the school level. After estimating the model, we retrieved Empirical Bayes estimates of school-level slopes and then examined correlations among these school “effects” for each scale.⁹ While all the school-level value-added estimates were positively correlated across score scales (mean and median correlations of .85 and .91), the strength of the correlation ranged considerably, from a low of .54 to a high of .99. The weakest correlation of .54 is found when scales calibrated under a hybrid approach using the 3PLM/GPCM combination with EAP estimation are compared with scales calibrated under a separate approach using the 1PLM/PCM combination with ML estimation. There is reason for optimism in the finding that the “main effect” choices of calibration approach, IRT model, and score estimation methodology seem to have little impact when the outcome of interest consists solely of normative comparisons among school-level value-added estimates. This optimism must be

tempered by the finding that the interaction of these choices can have a more noticeable impact.¹⁰

Betebenner (2009: this issue, pp. 42–51) presents an approach that allows one to assess the normative position of students in an achievement distribution and evaluate how this changes over time. While this is not a pure growth model, because it eschews the direct estimation of school-level quantities and instead focuses on student-level conditional achievement, neither does it fit cleanly within the criteria we have established for use of a value-added model. This is the approach Colorado is currently taking in the incorporation of a “growth model” into their statewide educational accountability system. For such uses a vertical score scale is not necessary. However, there is at least one clear drawback to normative approaches that eschews the need for a vertical scale. Namely, if all growth in a given state is relative, there is no guarantee that the state as a whole is making progress. It would be possible to have a situation where growth in aggregate student achievement is nonexistent or even decreasing, but this could be masked by a focus on the conditional growth percentiles of students relative to their peers. In addition, an approach such as Betebenner’s that directs focus at the student level are less amenable for the purposes of producing value-added school or teacher effect estimates with associated standard errors. Of course, given the present potential for value-added modeling to support misguided causal inferences, this may not be such a bad thing (Braun, 2005; Briggs & Wiley, 2008; Raudenbush, 2004; Rubin, Stuart, & Zannato, 2004).

Future Directions

One important limitation to this study related to the linking design of the data that were available to us from the CSAP reading assessment. The ideal check on the methods we have used in this study would involve the full panel data for Grades 3 to 7 between 2001 and 2006. While this data was not available to us for the present study, we have obtained it for use in a follow-up study. A limitation to the generalizability of our findings is that our analysis has only considered differences among vertical scales in the subject of reading. In the future we plan to examine vertical scales for

the CSAP math tests as well. A third limitation was our inability to conduct a fully concurrent calibration. This may have understated the potential differences in vertical scales as a function of calibration approach.

It is entirely possible that all of the vertical scales we have created in this study are to some extent biased. After all, we have no “true” scale against which such an evaluation can be made. The most likely source of bias would be a violation of the assumption that the construct of reading comprehension maintains a unidimensional interpretation over a multi-year grade span. It has previously been established that when tests measuring multiple dimensions are modeled using unidimensional methods, ability estimates will be biased (Ackerman, 1992; Béguin et al., 2000). The support for the unidimensionality assumption comes from factor analyses that indicate the presence of one strong single factor in each grade-level CSAP test (Colorado Department of Education, 2006). What remains unclear, and perhaps unlikely, is whether the single dimension identified in each grade represents the same thing. If the dimensional structure changes from test to test over time (what Martineau (2006) calls “construct shift”), the scores along a unidimensional vertical scale will be biased in the sense that they are not measuring the multidimensional construct of interest, but some sort of composite factor. If these scores are biased, it follows that growth projections will be biased as well. It is unclear at this stage whether bias due to multidimensionality would be exacerbated by any of the vertical scaling approaches taken in this study. Multidimensionality could be a possible explanation for the consistent but puzzling finding of decelerating growth (often dramatically decelerating growth) evident in almost all vertical scales that span the grades from elementary school through middle school. One possible solution would be to develop multidimensional vertical scales (Li & Lissitz, 2000; Oshima, Davey, & Lee, 2000; Reckase & Martineau, 2004), and this is an approach we are now currently investigating.

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Notes

¹The differences between growth models and value-added models can be subtle. One reason for this is the fact that when dummy variables linking teacher and/or schools to students are available, it becomes possible to estimate value-added quantities from a model that also parameterizes student-level growth. So, as we illustrate in a later section of this article, a hierarchical linear model can serve the same purpose as a value-added model through the inclusion of a third level of clustering and the subsequent estimation of empirical Bayes residuals at the teacher or school level. However, the assumptions and constraints imposed on random effect variance components when growth models are used to produce norm-referenced quantities can be quite restrictive relative to those imposed by more typical value-added models such as the EVAAS. A full discussion of these issues is outside the scope of this paper. For more information on value-added models see McCaffrey, Lockwood, Koretz, and Hamilton (2003) and McCaffrey, Lockwood, Koretz, Louis, and Hamilton (2004).

²Other designs that would in principle also support an IRT-based approach include what Holland and Dorans (2006, pp. 197–201) describe as the single group, equivalent-groups, and counterbalanced designs. These designs are much less commonly enacted in operational testing programs because they require multiple administrations of the same test, which can be both costly and create problems maintaining test security.

³Our use of the term linking is consistent with the terminology established by Mislevy (1992), Linn (1993) and Holland and Dorans (2006). Linking is more general than the term equating, which refers to the linking of scores on alternate forms of an assessment that are built to common content and statistical specifications. Holland and Dorans (2006) distinguish between three forms of linking: predicting, aligning and equating. Under their taxonomy, vertical scaling is a form of score aligning in which the underlying tests have similar constructs and reliability but differing difficulty and test-taking populations.

⁴For philosophical debates over the meaning of measurement in the context of IRT models, see Wilson (2004), Thissen and Wainer (2001), and Wright (1997).

⁵For an exception, see Thissen and Orlando (2001) for an approach in which item parameters are estimated using the 3PLM, but proficiency estimates are based upon summed scores. In this case, there would be a one-to-one relationship to raw scores and scale scores when using a more complex model.

⁶The patterns of the results we present here were consistent across both longitudinal cohorts. We present the results for just the first cohort due to space constraints.

⁷Interestingly, Ballou (2009) also found evidence of *increasing* within-grade SDs on NWEAs 2005 math tests for grades 2 to 9.

⁸For example, Hawaii and Oregon both proposed a hierarchical linear model as the basis for growth projections under the Growth Model Pilot Program.

⁹To simplify matters, in estimating these growth parameters, we restricted the sample to only those students who were present in the same school in grades 3, 4 or 5 for 2 out 3 years between 2003 and 2005. This reduced the available student sample for our analysis from 65,599 to 40,690. The students that were excluded represent those that either switched schools, moved to a private school, or left the state. For a different analysis in which the same data were used in the context of value-added modeling, see Briggs & Weeks (2009).

¹⁰Interestingly, the estimated correlation between school-level random effects for intercept and slope parameters appear to be very sensitive to the choice of scale. Under the 1PLM/PCM combination, this correlation is relatively strong and negative, regardless of whether EAPs or MLEs have been employed for the underlying scale. In contrast, under the 3PLM/GPCM combination, when EAPs are the basis for the underlying scale, the correlations drop dramatically.

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